Google Cloud

Data partitioning reduces cloud costs

Previously, you learned that database partitioning is the process of dividing data into separate data segments across different databases or data stores. Database Partitioning breaks your data into manageable segments so you can query datasets faster. You can also save on costs through database partitioning because data is stored in smaller segments, and less compute power is needed to query a smaller dataset. In this reading, you'll learn more about how database partitioning can save you on costs.

Cost of analytics

Traditionally speaking, data costs start with the cost of physical servers in an office. As data moves to the cloud, the costs shift to cloud storage or database instances. And as data stores grow, these costs increase. The two primary costs associated with data are cloud storage and compute power used to query.

Storage costs

The cost of cloud storage depends on the amount of data stored. Cloud storage services bill by the gigabyte (GB) stored. Costs also vary depending on the availability of the data. Storing data that needs to be accessed immediately, or in near real-time, will cost more than data that's in an archive state and accessed infrequently.

Query costs

Query costs will vary based on the database provider. Some providers will charge by the number of queries run, and the volume of data queried. Others might charge by hours that an instance is run, or the hardware size of the database. Generally, more compute is needed to query larger data sizes.

Cost savings with partitioning

Database partitioning can reduce data storage costs, and query costs for large organizations with years of data, or complex databases.

Partitioning and storage

Database partitioning allows you to distribute your data store across different storage types. Partitioned data can be stored in ways that place the data you need on a frequent basis in faster storage. Partitions with data that are no longer in active use, and are only saved for historical reference, can be archived and placed in low-cost storage.

Partitioning and querying

When you query a partitioned set of data, you'll not only have a faster response time, but you'll also pay for less query time and compute. The partitioned database will reduce the amount of data that needs to be scanned while running a query. Partitioned databases may also decrease the number of times that data will need to be scanned, which improves the query performance.

Key takeaways

Databases often require a lot of resources to store and process data. This is especially true for organizations that ingest large amounts of data daily, or organizations that have a substantial amount of historical data. Partitioning a database can cut down on both storage and compute costs incurred from processing and querying data.

Resources for more information

To learn more about the benefits of database partitioning, check out these links:

- Cockroach labs provides more information about database partitioning and how it can save on costs: https://www.cockroachlabs.com/blog/what-is-data-partitioning-and-how-to-do-it-right
- Timescale provides some best practices for data partitioning: https://www.timescale.com/learn/data-partitioning-what-it-is-and-why-it-matters